Pirate's Treasure*

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Abstract

Do countries that improve their protection of intellectual property rights gain access to new product varieties from technologically advanced countries? We build the first comprehensive matched firm level data set on exports and patents using confidential microdata from the US Census to address this question. Across several different estimation approaches we find evidence that these protections affect where US firms export.

Keyword*:* trade, innovation, intellectual property rights, patents

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1 Introduction

In recent years intellectual property rights (IPRs) have become a highly contentious policy issue, particularly with respect to international agreements such as the Trans-Pacific Partnership. In this paper we investigate how IPR policies affect the pattern of international trade through the extensive margin of the destinations to which firms export. We find that US firms that hold patents are more likely to export to countries with stronger IPR protections. This is found in the cross section as well as in looking at firms' responses to six different IPR reforms. As intellectual property rights have become increasingly central to production in the economies of modern industrialized countries, we hope that our work sheds light on an issue that is likely to grow in importance over time.

Our work is the first to match comprehensive information on patenting and exporting behavior at the firm level, and so we begin by documenting a number of new facts. Our unique data platform allows us to provide the most representative description of the nexus between trade and innovation to date. The results here follow in the vein of the highly influential work by Bernard, Jensen, and Lawrence (1995), who document that exporters tend to be larger, more productive, and pay higher wages than other firms, as well as Balasubramanian and Sivadasan (2011) who establish similar facts for patenting firms. Focusing on the manufacturing sector where these issues are the most important, we find a powerful relationship between trade and innovation. For example, only 9% of firms hold a patent but these producers account for 89% of all exports. On a similar score, patenting firms are much more likely to export relative to other firms and average exports increase substantially around the time of a firm's first patent. Firms that engage in both exporting and innovation are also significantly larger and more skill intensive than firms that only do one or the other.

We next estimate the relationship between the pattern of trade and intellectual property rights policies. The literature on this topic to date has primarily focused on the volume of trade and has delivered mixed results on the effects of a strengthening of IPRs. For example, Ferrantino (1993) finds no significant impact of better protections, Smith (1999) finds a negative impact, and Maskus and Penubarti (1995) and Ivus (2010) have found positive impacts. Given that firms that seek formal patent protection in the US could intuitively be more sensitive to these legal protections abroad than other businesses, we estimate whether firms with patents tend to export to countries where IPRs are better protected. Considering firm patenting in a gravity equation framework for the first time, we find that this is in fact the case.

Building on this approach, we consider how firms' export behavior responded to IPR reforms in six different countries. Given the panel nature of our data as well as large sample sizes, we are able to estimate the effects of these policy changes with a propensity score matching differences in differences approach. Firms with patents begin exporting more to these countries relative to comparable firms after reforms are passed in Argentina, Brazil, Colombia, the Philippines, and Turkey. We find no response to the reforms that were enacted in Venezuela shortly before the election of Hugo Chávez and what has been termed the "Bolivarian Revolution." Here, as well as in the gravity based estimations, we find that the effects are primarily due to responses in arm's length exports as opposed to related party

exports to firms' subsidiaries.

Our results suggest that IPR reforms should expand the range of goods exported to a country, which should improve the welfare of consumers through love of variety effects as well as help enhance the productivity of domestic firms (e.g., Hicks 1969, Krugman 1979, Goldberg, Khandelwal, Pavcnik, and Topalova 2010). In particular, Saggi (2012) considers a strengthening of IPRs in a North-South model with a northern monopolist who holds a patent. He finds that an improvement in protections in the South raises Southern welfare if and only if these reforms are necessary and sufficient to entice the monopolist to export to the South. Our work is the first to address this issue of the extensive margin of whether Northern firms' exports respond positively to reforms in the South.

In Section 2, we describe our data and document a number of new stylized facts on the relationship between trade and innovation. We then discuss our gravity based estimations in Section 3, our propensity score matching differences in differences estimations in Section 4, and conclude in the last section.

2 Data and Stylized Facts

2.1 Data

Our data come from a number of sources. For our estimations in this section as well as the next, we consider a sample of 158,200 firms and their export behavior to 112 countries, for a total of 17,722,200 observations.[1] Information on firms' exports comes from the Census Bureau's Longitudinal Firm Trade Transaction Database (LFTTD). It links export shipments reported by US Customs and Border Protection to individual firms beginning in 1992. For each transaction above $2,500, we have information on the value, destination country, port and a number of other variables. We recode shipments with a reported value below this threshold to zero for the sake of consistency. In order to get basic information on firm characteristics, we use the Longitudinal Business Database (LBD), which tracks the employment, payroll, state, and industry of every legally operating business establishment in the United States. These measures are aggregated to the level of the firm and matched to the LFTTD. We assign industry and state codes to businesses based on the sector in which they have the most employment, using payroll to break ties. Jarmin and Miranda (2002) provide a thorough description of the construction of the LBD as well as a number of associated descriptive analyses.

In order to obtain more detailed information on firm characteristics, we combine this merged data set with the Census of Manufacturers (CMF). This survey of all manufacturing establishments in the United States is conducted every five years ending in 2 or 7 (e.g., 1997). We only keep firms with positive employment that appear in both the LBD and CMF, drop those for which we have not been able to assign a Standard Industrial Classification (SIC)

[1] Throughout the paper we round the reported number of firms and observations as required under disclosure rules from the US Census.

code using the LBD data, and then drop those that are not determined to be primarily manufacturing firms. While this has the disadvantage of restricting our analysis to manufacturing, it allows us to account for a far greater level of firm heterogeneity and focuses the analysis on the sectors that matter the most for these issues. For example, Balasubramanian and Sivadasan (2011) document that manufacturing firms account for approximately 70% of all patents. Manufacturing also accounts for a substantial portion of US exports, suggesting that these industries are of primary importance for understanding the effects of intellectual property rights on trade.

To construct measures of firm level productivity, we draw upon the Annual Survey of Manufacturers (ASM). This survey is conducted each year between CMF years and allows us to follow plants over time. While the sampling frame is redone every five years, large producers are sampled with certainty and so we can follow them across the different five year intervals. We then use this sample to estimate Cobb Douglas production function parameters using the semiparametric methodologies developed by Levinsohn and Petrin (2003) and Wooldridge (2009). We do this by taking the data starting in 1982 up until 1997 and perform the estimations based on this sample. We then use these parameters to construct firm level productivity based on measures of value added, employment, and the capital stock from the 1997 CMF. The Solow residual is also calculated using information from the CMF.

Our information on patents comes from the US Patent and Trademark Office (USPTO) and covers the universe of granted patents in the US. Hall, Jaffe, and Trajtenberg (2001) describe these records in depth and Griliches (1990) discusses the large literature on the use of patents as indicators of innovation. Our records cover the period from 1975 to 2008, containing over 4 million granted patents. These data are merged onto the Longitudinal Business Database using a bridge originally constructed by and described in Kerr and Fu (2008), Balasubramanian and Sivadasan (2011), and Acemoglu, Akcigit, Bloom, and Kerr (2013). This bridge is developed using a matching algorithm between firm names and addresses contained in both the Census data and patent records. We further hand match a number of patents for some of the largest firms. We only keep patents with an assignee name and only keep private sector patents for the analysis. Following standard practice in the literature, we consider the date of a patent to be its application year instead of the year in which it was granted. This avoids issues of variable processing times between application and grant dates for different patents from the USPTO. Based on patent law, a firm's total patent stock is measured by aggregating all of the granted patents for which it has applied in the last 20 years.

In order to account for a number of foreign country characteristics, we use data from several additional sources. Our sample of countries is determined by our ability to obtain information on them. From Centre d'Etudes Prospectives et d'Informations Internationales (CEPII), for each country we obtain measures of: (i) distance from the US, (ii) whether or not the country shares a border with the US, (iii) an indicator for whether the country shares a common language with the US, (iv) the number of hours in time difference between the US and the country, (v) the area of the country in square kilometers, and (vi) whether the country shares a colonial relationship with the US.

Countries that are determined to have a colonial relationship with the US are Britain, France, Spain, and the Philippines. Distance is measured in kilometers following the great

circle formula and uses the latitudes and longitudes of the most populous city in each country. The indicator for countries that share a common language with the US equals one if a language is spoken by at least 9% of the population in both nations. Measures of GDP, GDP per capita, and the exchange rate are sourced from the World Bank. We complement these data with information from the Federal Reserve Bank of St. Louis and the CIA World Factbook for missing observations. From the data set developed by Helpman, Melitz, and Rubinstein (2008), we use measures of whether the country is an island and whether it is landlocked. We hand code the appropriate values for these two variables for Lithuania, the Slovak Republic, and Ukraine.[2]

To measure the strength of intellectual property rights protection across countries, we use an index (hereafter the "GP index") originally developed by Ginarte and Park (1997) and updated by Park (2008). This measure is constructed for every five years going back to 1960 and is based on five categories of patent protection: (i) the extent of coverage of patent laws, (ii) membership in international patent agreements, (iii) provisions for loss of protection, (iv) enforcement mechanisms, and (v) duration of protection. Each country receives a score between zero and one for each category, with a possible aggregate score of 5. In Table 1, we list the countries in our sample ranked in terms of their GP index in 1995. The rankings are intuitive, with industrialized nations like the UK and France at the top of the list and developing countries like Ethiopia, Mozambique, and Papua New Guinea at the bottom. It should be noted that while this list is not fully comprehensive, these countries account for more than 95 percent of all US exports (Lincoln and McCallum 2016).

2.2 Stylized Facts

In Tables 2 and 3 we use these data to document a number of new stylized facts with respect to innovation and exporting at the firm level. We find a strong link between the two. Here, as well as in the estimations in Section 3, we focus on the year 1997 in order to take advantage of the information on firm characteristics contained in the CMF which is not available from other sources. Wages are measured in thousands of dollars per employee and the skill share is defined as the percentage of workers in the firm who are not production workers. A few main stylized facts come out of these estimations:

1. Only 9% of manufacturing firms hold a patent but these businesses account for 89% of all exports
2. The vast majority of innovative firms are connected to the global economy
3. Exporters are significantly more likely to hold a patent than other firms
4. Innovative firms tend to sell to more countries when they export
5. Exporting firms tend to hold more patents when they hold at least one patent

[2] The CEPII data were developed by Head, Mayer, and Ries (2010). For more information on our measure of distance, see Mayer and Zignago (2011). Due to disclosure issues, the one country that is in all of our external data sets but is not in our sample is the Dominican Republic. We exclude this country to make the sample of countries the same across our different prior disclosure requests from the Census. The results throughout the paper are similar when including this country in the analysis.

6. Firm size, wages, and the skilled worker share increase as firms engage in exporting or patenting, with the set of firms that do both being exceptionally large.

The results in the last point in particular echo those from Bernard, Jensen, and Lawrence (1995) and Balasubramanian and Sivadasan (2011) in that exporters and patenting firms tend to be larger, pay better wages, and are more productive than other types of firms. The results for the group of businesses that both patent and export suggest that the main findings from the separate literatures on the characteristics of exporters and the characteristics of patenting firms are being driven the most by firms that engage in both activities.

In Figure 1 we look at the nexus between exporting and innovation across the firm employment size distribution. Both activities become more important as firm size increases, with the largest firms being especially important in terms of: (i) the percentage of exports coming from patenting firms and (ii) the percentage of exporters that have a patent. Figure 2 similarly shows significant variation across industries in exporting and innovation activities.[3] Here, we rank the sectors in ascending order of the percentage of exports coming from firms with a patent. The results make sense intuitively, with industries like Wood Products (SIC 24) having a low patent intensity of exports and those like Instruments (SIC 28) and Electronics (SIC 36) having a high share. Similar to Figure 1, we also plot the percentage of exporters that hold a patent as well as the related measure of the percentage of patenting firms that export. The correlations between each of the three measures in Figure 2 range from 59% to 69%.

In Figures 3 and 4 we consider the geography of innovative and exporting activities across the United States. Due to disclosure constraints that arise from cutting the data down to the state level in these maps, we focus on the value of exports coming from firms with a patent instead of the extensive margin of trade. This measure based on volumes generally tracks the extensive margin geographically, however; there is a 61% correlation across states between the patent intensity of exports based on volumes and the percentage of exporters that hold a patent. In Figure 3 we consider the total value of exports coming from patenting firms and in Figure 4 we consider the percentage of exports accounted for by these businesses.

A few conclusions come out of these figures. In terms of both total volumes as well as percentages, the Great Lakes region accounts for a significant amount of exports from patenting firms. Michigan, Ohio, and Illinois particularly stand out. When we move to considering patent intensity in Figure 4, some states such as California become less prominent. Others, such as Idaho with its specialization in science and technology, stand out even more. One notable state regionally is North Carolina, which has a number of high tech firms in the "Research Triangle" area around Raleigh, Durham, and Chapel Hill. Washington state, with its large information technology industry as well as direct access to the Pacific Ocean, is also prominent in both figures.

In order to get a sense of the link between patents and exports over time, Figure 5

[3]For the purposes of disclosure from the Census Bureau, like prior work we combine the Food and Kindred Products (SIC 20) and Tobacco Products (SIC 21) industries together due to sample size issues (e.g. Balasubramanian and Sivadasan 2011 and Lincoln and McCallum 2016). This is a common issue that researchers face when working with this data. For the same reasons, in Figures 3a and 3b we restrict the analysis to the continental US and consider Delaware, Maryland, and the District of Columbia to constitute a single state.

shows the average path of a firm's exports around the date of its very first patent. For this analysis, we restrict the sample to the firms that applied for their first granted patent in our sample period. We consider the five years before the date of the firm's first patent and the four years after. The sample thus includes a total of ten years, including the year in which the firm developed its initial patent. Our data extend from 1993 to 2006 and as such the analysis considers firms that first patented in the five year window between 1998 and 2002, incorporating more than 1700 firms.

The first series in Figure 5 shows that the average number of annual export transactions of a firm is roughly constant up until the year prior to its first patent application and then subsequently begins to rise. The second series shows a similar trajectory for the overall value of exports. We find that the nature of these trends are primarily driven by the top end of the distribution, however. Median values across firms for these measures tend to increase gradually over time around the date of a firm's first patent, as does the average number of countries to which firms export. We find similar patterns for both the mean and median values when limiting the sample to firms that exported in each of the ten sample years around their first patent. These results are in line with those of Balasubramanian and Sivadasan (2011), who find similar patterns with respect to a firm's size, and suggest that patenting is associated with real changes in exporting activity within firms.

In terms of foreign innovation, we find that this is also linked to exporting. Amongst firms with patents, only 9.5% hold a patent in which at least one inventor is located overseas, meaning that less than 1% of manufacturing firms overall have foreign patents. In terms of the number of inventors, the unconditional probability of an inventor on a given patent residing overseas for firms with patents is 6.3%. For non-exporters this figure is 1.8% and for exporters it is 6.9%, suggesting that exporters are much more likely to innovate abroad.

More generally, the stylized facts documented here point to significant heterogeneity in firms' exporting and patenting behavior. Intuitively, these differences amongst firms should lead to differential responses to foreign IPR policies. In the next sections we consider this empirically.

3 A Gravity Approach

In thinking about the effects of IPR policies on firms' behavior, a natural distinction to make in terms of which firms are the most dependent on IPR protection is whether or not a firm holds a patent. As they rely on legal protections in the US, it is possible that they are more dependent than other firms on legal protections abroad. In our estimation approach, we thus consider whether firms with patents are more sensitive to foreign IPR policies than others in terms of their export behavior. If we assume that IPR policies primarily affect companies that are reliant on patents and that stronger IPR protections are associated with an increased likelihood of exporting by firms with patents relative to others, it would suggest that these policies have extensive margin effects on what types of goods are ultimately imported into

these markets.[4]

There are a number of mechanisms that may be at play in deterring firms from exporting to countries where their technology is more likely to be pirated. For example, if the probability of imitation is high, the expected profits from selling in a country may not be large enough to cover the up front costs of beginning to export to that market. If a firm's technology is pirated in a given country, this may also erode its monopoly power in other markets despite the enforcement mechanisms that have been put in place internationally to try to prevent this from happening. While applying for a patent makes the details of a firm's invention public, simply reading through the technical details of a large number of inventions is likely to leave firms uncertain as to which products are the most likely to be profitable to imitate. One potential reason for why firms might be deterred from exporting to a country with weak IPRs is that the popularity of their products selling well could provide a valuable signal to foreign firms that imitation is worthwhile, raising the probability of piracy and the erosion of profits that could come with it.

Motivated by the gravity approach that has been widely used to estimate the determinants of international trade flows, to consider this empirically we use the fixed effects specification:

$$Export_{ic} = \mu_i + \mu_c + \beta \cdot Pat_i \cdot GP_c + \gamma \cdot X_{ic}^{grav} + \varepsilon_{ic}.$$

Here $Export_{ic}$ is an indicator for whether firm i exported to country c. The fixed effects μ_i account for a variety of firm characteristics, such as productivity, size, and patenting status. The fixed effects μ_c control for country characteristics such as distance to the US, GDP per capita, and the protection of intellectual property rights. X_{ic}^{grav} contains a number of controls discussed shortly. Our main object of interest here is the coefficient β on the interaction term between the indicator for whether firm i holds a patent Pat_i and the GP index for country c, which we denote GP_c. For ease of interpretation, we normalize this index to lie between zero and one. If β is estimated to be different than zero, this would suggest that there is a relationship between a firm's dependence on patent protection and its choice of where to export.

In column (1) of Table 4 we consider this specification for overall export status. All estimations cluster standard errors at the level of the firm. We use the GP index from 1995 in these estimations, which is the closest date in which it is available relative to our sample year of 1997. In addition to our main interaction term, we include interactions between: (i) the log employment of the firm and the GP index of the country, (ii) the log of average firm wages and the GP index, (iii) the skill share of the firm and the GP index, and (iv) the productivity of the firm and the GP index. As above, the skill share of the firm is measured as the percentage of non-production workers in total employment and productivity is measured using the semiparametric approach of Levinsohn and Petrin (2003).

[4]For related work on the issues of IPR protections, innovation, and trade, see Arkolakis, Ramondo, Rodríguez-Clare, and Yeaple (2014), Atkeson and Burstein (2010), Aw, Roberts, and Xu (2011), Bilir (2014), Branstetter and Sakakibara (2001), Branstetter and Saggi (2011), Delgado, Kyle, and McGahan (2013), Furman, Porter, and Stern (2002), Glass and Saggi (2002), Grossman and Lai (2004), Hallak (2006), Ivus (2011, 2015), Javorcik (2004), Keller (2004), Keller and Yeaple (2013), Lai (1998), Lerner (2002), Murray and Stern (2007), and Williams (2013).

In X_{ic}^{grav} we also include a set of additional interactions between our patent indicator Pat_i and a number of the country characteristics that are widely used to explain trade patterns in the gravity equation literature. This includes: (i) the GDP per capita of the country, (ii) the exchange rate for the country, measured in log foreign currency units per dollar, (iii) log distance from the US, (iv) whether or not the country shares a border with the US, (v) whether or not the country shares a common language with the US, (vi) the geographic area of the country in square kilometers, (vii) the number of hours in time difference between the US and the country, (viii) an indicator for whether the country shares a colonial relationship with the US, (ix) an indicator for whether the country is an island, and (x) an indicator for whether the country is landlocked. The results on these interaction terms are of interest more generally, as this is the first analysis to our knowledge to consider how firm patenting relates to the pattern of trade in a gravity framework.

The estimate of $\hat{\beta}$ in our baseline specification suggests that an improvement in IPR protections from the lowest to the highest possible GP index score is associated with a 2.9% increase in the probability that a firm with a patent exports to the country relative to an otherwise similar firm without patents, ceteris paribus. Thus, if a country improved its level of IPR protection from that of Angola to that of France, for example, this would be associated with a relative 2.1% increase in the probability of a firm exporting to the country. This estimate is strongly statistically significant, with a t statistic of 37. These results are consistent with the idea that IPR policies affect the composition of exports through the extensive margin of trade.

Statistically significant coefficient estimates on the other interaction terms imply that firms with a larger, better paid, and a more highly skilled workforce are more likely to export to markets that better protect intellectual property rights. Patenting firms are more likely to sell to wealthier markets, those that share a common language with the US, and those that had a colonial relationship with the US. They are also more likely to export to markets that are further away, with the important exception of countries that share a border with the US. Interestingly, firms with patents do not seem to be particularly sensitive to exchange rates. If firms price to market, exchange rate pass through is incomplete, and patents afford firms with market power that is disproportionately affected by incomplete pass through, one could imagine obtaining the converse result. This suggests that exchange rates do not have a large effect on the technology composition of the firms exporting to a particular country.

Differences in time zones have a statistically significant effect on firms with patents relative to others, even when controlling for distance between the two countries, but the effect is not large. This is also perhaps surprising; it would not be unreasonable to think that technologically sophisticated exports could be significantly affected by the communication costs imposed by time zone differences. For example, Stein and Daude (2007) have found significant effects of time zone differences on foreign direct investment flows.

In the next column of Table 4, we consider an additional regressor. Firms with more newly developed patents are likely to have more monopoly power and thus could potentially be more sensitive to IPR protections than other firms. We thus consider specifications similar to those found in the first column but additionally include a triple interaction term between: (i) whether or not a firm has a patent, (ii) the GP index for the country, and (iii) the average age of a firm's patents. Our analysis here has an advantage over the literature on technology

obsolescence in that we can develop a firm level measure of how cutting edge an individual firm's technologies are, as opposed to relying on broader industry measures.

The negative and statistically significant coefficient on this term suggests that firms with more recently developed products tend to be more sensitive to IPR policies. The coefficient on our primary regressor increases somewhat with the inclusion of the additional triple interaction term. The coefficients on the other independent variables remain similar to those of their counterparts in column (1). As the average age of a firm's patents is 6.8 years, these results suggest that a firm with a patent stock of average age would be 1.4 percentage points less likely to export to a high IPR destination than a firm whose patents were all newly developed. Intuitively, the estimates suggest that a firm whose patents were all about to expire, on the other hand, would not be predicted to be sensitive to IPR protections relative to a firm without patents.

We consider a number of robustness checks on these results. To deal with the concern about collinearity bias in the estimator of Levinsohn and Petrin (2003) raised by Ackerberg, Caves, and Frazer (2015), we estimate the specification using the approach of Wooldridge (2009), finding similar results. The same is true when just using a simple Solow residual, which does not require estimating production function parameters from the ASM. Controlling for GDP rather than GDP per capita also yields similar results.

We further consider estimations that drop the interaction terms between the GP index and firm productivity, wages, and the share of skilled workers. We include these controls in our main reported regressions to ensure that we are not capturing the effects of these alternate factors. However, including them potentially partials out at least some of the variation that we care about. When excluding these controls in the baseline specification in column (1), we obtain an estimate of β on our main interaction term of 4.0%. This is somewhat higher than our initial estimate of 2.9% but of a similar magnitude.

One concern in working with the LFTTD is that information on exports to Canada is collected separately from that of other countries. This is due to agreements between the two countries intended to reduce reporting burdens on firms. Exports to other countries can be easily matched to firms using the Employer Identification Number found in both data sets. Shipments to Canada, however, have to be identified by first performing an automated matching algorithm between firm names in the LBD and those listed on the export transactions in the US Customs data. A hand matching process is then conducted for non-matched high value exporters. This imperfect approach induces some measurement error into the data on exports to Canada, particularly with respect to the extensive margin of trade. This is not an issue for Mexico, the United States' other major trading partner under the North American Free Trade Agreement. Given these concerns, we estimate the specification dropping Canada from the analysis. This gives very similar results to the baseline estimations found in Table 4.

We also consider estimating the specification above for the years 1993 and 2002 in order to see whether the results would hold over time. Given the typical time lag for granting patents by the USPTO and the fact that the LFTTD data only starts in 1992, these are the only other years that we could consider where we have access to the CMF as well as a well measured firm patent stock. Since the data on exports to Canada are especially poorly measured for the year 1992, we consider the year 1993 and use information on firm characteristics from the

1992 CMF. These estimations are thus limited to firms that were active in both years. We also find significant effects in 1993 and 2002. These results are robust to dropping Canada from the analysis as well.

In order to explore the effect of intellectual property rights protection on the decision of where to export in further depth, we leverage additional information on the nature of firms' trading partners. The LFTTD contains information on whether each export transaction is between "related parties" or not. Here, an exchange between related parties is defined as a shipment between a company and one of its subsidiaries abroad. We similarly define arm's length exports as shipments that are not between related parties. Given that these subsidiaries have the same incentive to not let the intellectual property of the firm be pirated, we would expect there to be a significantly weaker relationship between the destination of related party exports and IPR protection. We do not necessarily expect this relationship to be zero, however, as IPR policies could affect where firms have subsidiaries in the first place. Aggregating firms' arm's length and related party transactions separately, we can identify whether a firm exported to a given country in each particular way.

To give some context to these estimations, 23% of exporters have related party exports. This figure is 43% for patenting firms and 16% for firms without a patent. For businesses that have related party exports, these shipments on average account for 24% of total exports. For firms with patents this share is lower at 21% than for those that do not hold a patent, for which the share is 27%. Thus, while patenting firms are more likely to have related party exports, conditional on having related party exports, those that do not hold a patent tend to send a greater share of their foreign shipments to affiliates abroad. Finally, when looking at the share of related party exports that come from firms with patents, the figure is even higher than that for overall exports. Fully 95% of related party exports come from firms that patent.

In columns 3-6 of Table 4 we report our findings for arm's length and related party exports separately. Each set of estimations follows the analysis in columns 1 and 2. We find significant differences for the two types of exports. Point estimates for our main coefficient in the arm's length estimations are quite similar to those for overall exports. This is consistent with the fact that having related party exports but not having arm's length exports to a given country is relatively rare. In the estimations for related party exports, however, the coefficients decline in magnitude significantly, suggesting a much weaker relationship with the coefficient dropping from 2.9% to 0.4%. The estimates maintain the same sign across all four columns, potentially indicating a similar mechanism at play. These results make sense intuitively in that the probability of piracy resulting from related party transactions is likely significantly lower than that for arm's length exports. We take these results to be additional confirmation that concerns over piracy are what is driving our main findings.

4 Evidence From Six IPR Reforms

While the estimations described in the previous section use a wealth of information on firm and country characteristics across a large number of markets, they do not take advantage of the panel nature of our data. Here we consider how firms' export behavior changed over time in response to six different intellectual property rights reforms in foreign countries during our sample period. Given our large sample size, we are able to consider the effects of these one by one, rather than pooling the data on all of the reforms as has been done in the past in looking at the effects of these policy changes on other outcomes. This allows us to consider the effects of strengthening IPR protections in settings with very different policy and political environments as well as varied market structures. Our data preparation here is similar to that described in Section 2, with the exception that we consider variation over time and do not restrict the sample to firms that appear in the CMF. To be consistent with the previous estimations, however, here we focus on the manufacturing sector as well.

One issue with our estimations in Section 3 in particular is that while the Ginarte and Park index has a long history of being used in the literature as a measure of intellectual property rights protection, it may be measured with error. If this measurement error is classical it would have the effect of biasing our results towards finding no effect, but this is difficult to assess. In looking at a set of IPR changes in countries where other sources suggest that de jure shifts in legal protections were accompanied by de facto changes in actual protections, we are able to address this concern.

The set of reforms that we consider come from the pioneering work of Branstetter, Fisman, and Foley (2006) and Branstetter, Fisman, Foley, and Saggi (2011). These reforms were initially identified by considering the lists of reforms in Maskus (2000) and Qian (2007) as well as by looking at the set of countries for which the GP index showed a significant strengthening in IPR protection over time. The reforms in Maskus (2000) in particular were identified in part by going through several years of the US Trade Representative's publication *National Trade Estimates Report on Foreign Trade Barriers*. Taken together, these reforms have been shown to have had real consequences on international technology transfer within US multinationals as well as in spurring industrial development in the reforming countries. An extensive description of these reforms is found in the original study by Branstetter, Fisman, and Foley (2006).

As our export data begin in 1992 and the patent data begin to show attrition for patent applications after 2002 due to lags in processing from the US Patent and Trademark Office, we are limited to a subset of the reforms that have been used in the prior literature. In particular, we focus on reforms that were enacted in 1994 or afterwards. This is in order to observe firm export behavior for at least two years prior to the policy change and avoid the attenuation bias that could result from anticipatory behavior. When possible, we choose the pre-period for our analysis to be three years prior to the reform. The post-period is always chosen to be four years afterwards.[5] This leads us to focus on the reforms in the countries

[5]In constructing the data samples, since for Brazil and the Philippines as well as for Colombia and Venezuela the pre-period and post-period years are the same, we take extra steps to ensure that no confidential information is disclosed from the Census. For these sets of countries we first build an initial data set and then

listed in Table 5.

Like much of the literature on the effects of these reforms we use a differences in differences estimation strategy, considering whether firms with patents differentially began exporting to the market under consideration after the reform. Given the large sample sizes for each country noted in Table 5, we are able to combine this approach with propensity score matching techniques as developed in Heckman, Ichimura, and Todd (1997) and described in Blundell and Dias (2009). This ensures that in doing the differences in differences estimations the effective matched comparison group of non-patenting firms is similar to patenting firms before the IPR reform is enacted. This methodology has been shown to be effective in identifying impacts when conducting a randomized social experiment is infeasible, as is the case when considering the effects of IPR reforms.

In mapping our estimations to the language of the microeconometrics literature, our treatment group comprises firms that hold a patent and the control group includes all other firms. The "treatment" is the IPR reform in the country and the outcome variable is a binary indicator for whether the firm exported to the country or not in a given year. More formally, our estimates are given by:

$$\hat{\alpha}_{MatchingDD} = \frac{1}{n_1} \sum_{i \in I_1 \cap S_p} \left\{ (Y_{it_1} - Y_{it_0}) - \sum_{j \in I_0 \cap S_p} \omega_{ij} \left(Y_{jt_1} - Y_{jt_0} \right) \right\}.$$

Here I_1 denotes the treatment group of firms that hold a patent, I_0 denotes the control group of other firms, S_p represents the common support, t_1 is the post-period, and t_0 is the pre-period. Y is an indicator for export status and n_1 is the number of firms in the treatment group that are also in the common support $i \in I_1 \cap S_p$. The weights ω_{ij} are constructed from the estimation of the propensity score. Our matching procedure is done in the pre-period year prior to the reform.

In doing the estimations, we limited the matching procedure for each firm i to those other businesses that were in the same 2 digit SIC industry,[6] implicitly accounting for sectoral characteristics such as product life cycle length as well as differences in the ease of imitation. Within this set of firms, we match on log employment, log wages, percentage employment growth prior to the start of the sample, percentage real wage growth prior to the start of the sample, and a set of region dummy variables. Employment and payroll growth are calculated over the three years prior to the beginning of the sample using the LBD. Importantly for combining this with a differences in differences framework, matching on these variables in the pre-period ensures that we are not comparing firms that initially have fundamentally different growth trends.

In our baseline estimations we use an Epanechnikov kernel matching approach with a bandwidth parameter of .05 and impose a common support. Following the influential work of Dehejia and Wahba (1999, 2002), we use a logit specification but the results are robust to this choice. As we do the propensity score estimations within industry, matching on state

restrict each sample to contain the intersection of the two different samples. This affected few observations.

[6] Note that, like above, in all of the matching estimations we combine the 2 digit SIC industries 20 and 21 together due to sample size issues. The approach allows for a sufficiently large comparison group in the matching estimations, since they are done within industry.

indicators sometimes cuts the data too thinly and so we choose to match on region instead. This is based on the nine official Census Regions corresponding to the areas of New England (1), the Middle Atlantic (2), the Great Lakes (3), the Prairie Midwest (4), the South Atlantic (5), the South Central East (6), the South Central West (7), the Mountain West (8), and the West Coast (9). As before, we use employment counts for each firm to determine its primary region, with payroll used to break ties. Throughout the analysis, wages are deflated with the consumer price index. Balancing tests suggest that the matching procedure generally works well.

Standard errors are calculated using bootstrap techniques with 100% resampling. In doing the estimations, the entire matching and differences in differences procedure is bootstrapped. This allows the process to capture the components of the variance in the estimates that are due to the estimation of the matching itself. Note that while bootstrapping methods have been shown to be invalid for nearest neighbor matching, they can be applied in this context in using kernel and local linear matching estimators (Abadie and Imbens 2008). Given our sample size, the bootstrap procedure is computationally intensive as we have to perform the within industry matching procedure for each sample. We thus use 100 replications for each of the estimations of the standard errors. This gives similar answers to estimations with 50 replications.[7]

Table 6 presents the results. We consider the effects of these reforms on three types of exports across six different countries for a total of eighteen estimations. Focusing on total exports to begin, we find statistically significant results for the effects of the reforms in all but one of the countries. The significant effects are broadly consistent and are the largest for Brazil and the smallest for Colombia, ranging from 1.1 to 2.7%. The magnitude of these estimates is broadly consistent with our cross sectional results in Section 3 and suggests that these policy changes had real effects on US firms' export behavior. The p-values on the coefficients for exports to these five countries are all below 2%.

For Venezuela we find small and statistically insignificant effects. As with all of the other estimations, here we choose our post-period to be four years after the reform, which is 1998. That year saw significant policy uncertainty in Venezuela, with a decline in confidence in the existing parties, the election of Hugo Chávez (who had previously been involved in a failed coup attempt in 1992), and the beginning of what has been termed as the "Bolivarian Revolution." Policy uncertainty has been shown to have significant impacts on international trade (e.g., Handley and Limão 2014, 2017) and we think that this is likely playing an important role in our findings.

These results are robust to a range of matching approaches. This includes: (i) using a wider (.1) or narrower (.01) Epanechnikov kernel bandwidth parameter, (ii) using a Gaussian kernel, (iii) using local linear regression, (iv) using a probit rather than a logit specification to estimate the propensity score, and (v) including higher order terms of the covariates on firm operations. In the last robustness exercise in particular, following the suggestions in Dehejia and Wahba (1999, 2002) we include log employment squared, log wages squared, the interaction between log employment and log wages, pre-period employment growth squared,

[7]Bootstrap estimations are done on the observations from the common support in the baseline estimations so that the two sets of estimates are based on the same set of observations. This ensures that the same sample is used so that there are no issues with disclosure from the US Census. This is not a very restrictive condition.

pre-period wage growth squared, and the interaction between pre-period employment growth and pre-period wage growth.

A final robustness check has to do with the performance of the baseline matching procedure itself. Employment and wage growth typically match exactly throughout the propensity score distribution. Levels of employment and wages also tend to be well balanced but with two exceptions. These variables prove to be difficult to match exactly in the bottom two deciles of the propensity score distribution for employment and the bottom decile for wages regardless of the many approaches suggested in the literature. Average values for these variables in the bottom two deciles tend to be similar, although the differences are usually statistically significant. These firms are those that are predicted to have a low probability of holding a patent based on the covariates and are thus given a low weight in our estimations since they are poor matches. As a robustness check, we run the matching differences in differences estimations dropping firms in either the lowest or two lowest propensity score deciles. Results for all countries are robust to these alternative approaches.

Following on our analysis in the last section, we then brake these estimations down to consider the reforms' effects on arm's length exports and related party exports separately. As before, although we do not necessarily expect the effects to be zero for related party exports, we expect the estimates to be smaller since firms' subsidiaries have incentives to protect the intellectual property of the firm. We find similar results as in Section 3. The estimates for arm's length exports are all comparable to those for total exports. The estimates for related party exports, in contrast, are typically much smaller. For two countries the coefficient for related party exports is slightly less than half of the estimate for arm's length exports and is statistically significant. For the other four reforms the estimated coefficient is even smaller and statistically insignificant. These results are robust to the same checks conducted for total exports.

5 Conclusion

In this study we have explored how intellectual property rights policies affect the pattern of international trade. We began by documenting a number of new stylized facts on firm trade and innovation, leveraging our access to a confidential database covering all US businesses. Using different estimation approaches, we then found support for a significant role for IPR policies in determining where firms export.

Our work has a number of implications. It suggests that intellectual property rights protections affect the distribution of goods available to both consumers and producers in foreign countries. This is likely to have impacts on both groups due to love of variety effects and access to a wider range of intermediate inputs. It is particularly important in the developing country context, where firms are often reliant on imports of intermediate capital goods that embody the latest technologies (Eaton and Kortum 2001, Burstein, Cravino, and Vogel 2013). The results also help explain why IPRs have become such an important policy

priority for the US Trade Representative in agreements; exports from the United States are overwhelmingly dominated by innovative firms. Officials in other industrialized countries have made these issues a top priority as well, suggesting that a similar dynamic may be at play in these markets.

Moving forward, a variety of questions remain to be addressed. Understanding the tradeoff between the benefits from gaining access to a greater number of imported goods on one hand and the benefits of lower prices due to imitation on the other hand is a first order issue. How the mechanism found here mediates the effects of trade on technology diffusion and the international convergence in living standards is also yet to be fully understood. As the rate of technological progress and the associated importance of intellectual property rights increases, these issues promise to only grow in importance with the passage of time.

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Table 1: Intellectual Property Rights Protections of the Countries in Our Sample

	Country	GP		Country	GP		Country	GP
1	Belgium	4.54	39	Jamaica	2.86	77	Senegal	1.98
2	Denmark	4.54	40	Nigeria	2.86	78	Togo	1.98
3	France	4.54	41	Uganda	2.85	79	Rwanda	1.95
4	Netherlands	4.54	42	Ghana	2.83	80	Mauritius	1.93
5	UK	4.54	43	Venezuela	2.82	81	Iran	1.91
6	Finland	4.42	44	Cyprus	2.78	82	Congo	1.90
7	Japan	4.42	45	Algeria	2.74	83	Honduras	1.90
8	Sweden	4.42	46	Colombia	2.74	84	Ivory Coast	1.90
9	Canada	4.34	47	Argentina	2.73	85	Bangladesh	1.87
10	Italy	4.33	48	Peru	2.73	86	Syria	1.87
11	Austria	4.21	49	Malaysia	2.70	87	Madagascar	1.85
12	Spain	4.21	50	Lithuania	2.69	88	Saudi Arabia	1.83
13	Switzerland	4.21	51	Iceland	2.68	89	Nepal	1.79
14	Australia	4.17	52	Turkey	2.65	90	Benin	1.78
15	Germany	4.17	53	Sudan	2.61	91	Chad	1.78
16	Ireland	4.14	54	Haiti	2.58	92	Niger	1.78
17	Hungary	4.04	55	Philippines	2.56	93	Egypt	1.73
18	New Zealand	4.01	56	Sierra Leone	2.45	94	Tunisia	1.65
19	Chile	3.91	57	Kenya	2.43	95	Zambia	1.62
20	South Korea	3.89	58	Thailand	2.41	96	Malta	1.60
21	Norway	3.88	59	Bolivia	2.37	97	Costa Rica	1.56
22	Singapore	3.88	60	Trinidad	2.33	98	Indonesia	1.56
23	Ukraine	3.68	61	Tanzania	2.32	99	Paraguay	1.53
24	Russia	3.48	62	Zimbabwe	2.28	100	Brazil	1.48
25	Greece	3.47	63	Fiji	2.20	101	Panama	1.46
26	Poland	3.46	64	China	2.12	102	Pakistan	1.38
27	South Africa	3.39	65	Iraq	2.12	103	India	1.23
28	Portugal	3.35	66	Liberia	2.11	104	Guyana	1.13
29	Bulgaria	3.23	67	Cameroon	2.10	105	Nicaragua	1.12
30	El Salvador	3.23	68	Gabon	2.10	106	Guatemala	1.08
31	Taiwan	3.17	69	Uruguay	2.07	107	Jordan	1.08
32	Israel	3.14	70	Ecuador	2.04	108	Angola	0.88
33	Mexico	3.14	71	Malawi	2.03	109	Burma	0.20
34	Sri Lanka	2.98	72	Somalia	2.00	110	Ethiopia	0.00
35	Czech Rep.	2.96	73	B Faso	1.98	111	Mozambique	0.00
36	Slovak Rep.	2.96	74	CA Repub	1.98	112	PN Guinea	0.00
37	Hong Kong	2.90	75	Mali	1.98			
38	Vietnam	2.90	76	Mauritania	1.98			

Notes: The table reports the countries in our sample in Sections 2 and 3 along with the associated intellectual property rights ("GP") index in 1995 for each nation from Park (2008). "UK" refers to the United Kingdom, "Trinidad" refers to both Trinidad and Tobago, "B Faso" refers to Burkina Faso, "CA Repub" refers to the Central African Republic, and "PN Guinea" refers to Papua New Guinea. The ordering makes sense intuitively, with countries like the UK and France at the top of the list and Ethiopia, Mozambique, and Papua New Guinea at the bottom.

Table 2: Exporting and Innovation Across Firms

	Mean	Standard Deviation
Firms That Export (pct)	29	45
Amongst Patenting Firms	82	38
Amongst Non-Patenting Firms	24	43
Firms That Hold a Patent (pct)	9	28
Amongst Firms That Export	25	43
Amongst Firms That Do Not Export	2	15
Amongst Exporters, Average Number of Destinations	5	8
Amongst Patenting Firms	11	13
Amongst Non-Patenting Firms	3	5
Amongst Patenting Firms, Average Number of Patents	30	368
Amongst Firms That Export	36	406
Amongst Firms That Do Not Export	2	6

Notes: The table presents a number of stylized facts on the nexus between exporting and innovation for US manufacturing firms. They are based on the 158,200 firms in 1997 from the sample described in Section 2 of the paper. Exporting is associated with larger levels of patenting and vice versa.

Table 3: Firm Characteristics

	Mean	Standard Deviation
Firms That Do Not Patent Or Export		
Employment	28	69
Wages	29	25
Skill Share	24	18
Log Productivity	3.05	0.83
Firms That Patent But Do Not Export		
Employment	79	249
Wages	38	51
Skill Share	32	20
Log Productivity	3.02	0.86
Firms that Export But Do Not Patent		
Employment	113	550
Wages	35	30
Skill Share	30	19
Log Productivity	3.10	0.87
Firms That Patent And Export		
Employment	947	5851
Wages	40	26
Skill Share	37	20
Log Productivity	3.09	0.79

Notes: The table describes the mean and standard deviation of the firm specific covariates included in our regressions in Section 3. We consider four categories of firms, based on whether they export or not and whether they hold a patent or not. The figures consider the same sample as in Table 2. Employment is measured in thousands of workers, wages are measured in thousands of dollars per employee, and the skill share is defined as the percentage of workers in the firm who are not production workers. We see a clear link between exporting and patenting when looking across the four categories, with firms engaged in both activities being dramatically larger than those in any of the other groups.

Table 4: Patenting in a Gravity Framework

	Dependent Variable is a (0,1) Indicator Variable For Whether the Firm Exports To The Country					
	Total Exports		Arm's Length		Related Party	
(0,1) Patent * GP Index	0.0291	0.0432	0.0248	0.0425	0.0038	0.0067
	(0.0014)	(0.0036)	(0.0014)	(0.0036)	(0.0005)	(0.0014)
(0,1) Patent * GP Index * Age		-0.0021		-0.0021		-0.0004
		(0.0005)		(0.0005)		(0.0002)
Log(Employment) * GP Index	0.0354	0.0354	0.0348	0.0348	0.0113	0.0113
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0003)	(0.0003)
Log(Wages) * GP Index	0.0146	0.0146	0.0144	0.0143	0.0016	0.0016
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0001)	(0.0001)
Skill Share * GP Index	0.0649	0.0646	0.0638	0.0636	0.0059	0.0059
	(0.0021)	(0.0021)	(0.0021)	(0.0021)	(0.0008)	(0.0008)
Productivity * GP Index	-0.0006	-0.0006	-0.0005	-0.0006	-0.0002	-0.0003
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0001)	(0.0001)
(0,1) Patent * Log(GDP p.c.)	0.0268	0.0268	0.0265	0.0265	0.0047	0.0047
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0001)	(0.0001)
(0,1) Patent * Log(Exch. Rate)	-0.0004	-0.0004	-0.0003	-0.0003	-0.0002	-0.0002
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)
(0,1) Patent * Log(Distance)	0.0442	0.0442	0.0434	0.0434	0.0032	0.0032
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0005)	(0.0005)
(0,1) Patent * Shares Border	0.2478	0.2478	0.2454	0.2454	0.0885	0.0885
	(0.0028)	(0.0028)	(0.0028)	(0.0028)	(0.0019)	(0.0019)
(0,1) Patent * Same Language	0.0110	0.0110	0.0112	0.0112	0.0017	0.0017
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0002)	(0.0002)
(0,1) Patent * Area	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
(0,1) Patent * Time Difference	-0.0048	-0.0048	-0.0047	-0.0047	-0.0002	-0.0002
	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0001)
(0,1) Patent * Colony	0.1011	0.1011	0.0991	0.0991	0.0292	0.0292
	(0.0016)	(0.0016)	(0.0016)	(0.0016)	(0.0009)	(0.0009)
(0,1) Patent * Island	-0.0157	-0.0157	-0.0155	-0.0155	0.0022	0.0022
	(0.0007)	(0.0007)	(0.0007)	(0.0007)	(0.0003)	(0.0003)
(0,1) Patent * Landlocked	-0.0126	-0.0126	-0.0125	-0.0125	-0.0014	-0.0014
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0002)	(0.0002)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Estimations consider the determinants of exporting to a particular country. All specifications cluster standard errors at the level of the firm. Adjusted R^2 values are close to .24 in columns (1)-(4) and are close to .14 in columns (5)-(6). The variables "Patent", "Shares Border", "Common Language", "Colony", "Island", and "Landlocked" are all indicator variables. "GDP p.c." refers to GDP per capita. There are 158,200 firms in the sample and we consider the determinants of their export behavior to a set of 112 countries in 1997, for a total number of 17,722,200 observations. These countries are listed in Table 1.

Table 5: IPR Reforms

Country	Reform Year	Pre-Period	Post-Period	Observations
Argentina	1996	1993	2000	270
Brazil	1997	1994	2001	273
Colombia	1994	1992	1998	283
Philippines	1997	1994	2001	273
Turkey	1995	1992	1999	266
Venezuela	1994	1992	1998	283

Notes: The table considers the reforms for our propensity score matching differences in differences analysis. The pre-period is chosen to be two to three years prior to the reform based on data availability. The post-period is always chosen to be four years after the reform. The number of observations is reported in thousands.

Table 6: Propensity Score Matching Differences in Differences Estimations

Country	Coefficient	Standard Error	t Statistic
Argentina			
Total Exports	0.02028	(0.00576)	3.52
Arm's Length Exports	0.01987	(0.00580)	3.42
Related Party Exports	0.00952	(0.00283)	3.37
Brazil			
Total Exports	0.02661	(0.00677)	3.93
Arm's Length Exports	0.02660	(0.00680)	3.91
Related Party Exports	0.01010	(0.00354)	2.85
Colombia			
Total Exports	0.01053	(0.00440)	2.40
Arm's Length Exports	0.00820	(0.00442)	1.85
Related Party Exports	0.00033	(0.00244)	0.14
Philippines			
Total Exports	0.01244	(0.00476)	2.61
Arm's Length Exports	0.01270	(0.00477)	2.66
Related Party Exports	0.00378	(0.00296)	1.28
Turkey			
Total Exports	0.01944	(0.00498)	3.90
Arm's Length Exports	0.02031	(0.00506)	4.01
Related Party Exports	0.00055	(0.00168)	0.33
Venezuela			
Total Exports	0.00400	(0.00565)	0.71
Arm's Length Exports	0.00368	(0.00570)	0.65
Related Party Exports	0.00108	(0.00234)	0.46

Notes: The table reports propensity score matching differences in differences estimations for 6 countries and 3 different types of exports, for 18 estimations total. Standard errors are calculated using bootstrap methods with 100% resampling and 100 replications used for each estimation. Here the treatment group are firms in the US that hold a patent and the treatment itself is the IPR reform. We find statistically sigfnificant effects on total exports for the reforms in five out of the six countries, with changes in Venezuela being the exception. The estimated effects are systematically larger for arm's length exports than for related party exports.

Figure 1: Exporting and Patenting Across The Firm Size Distribution

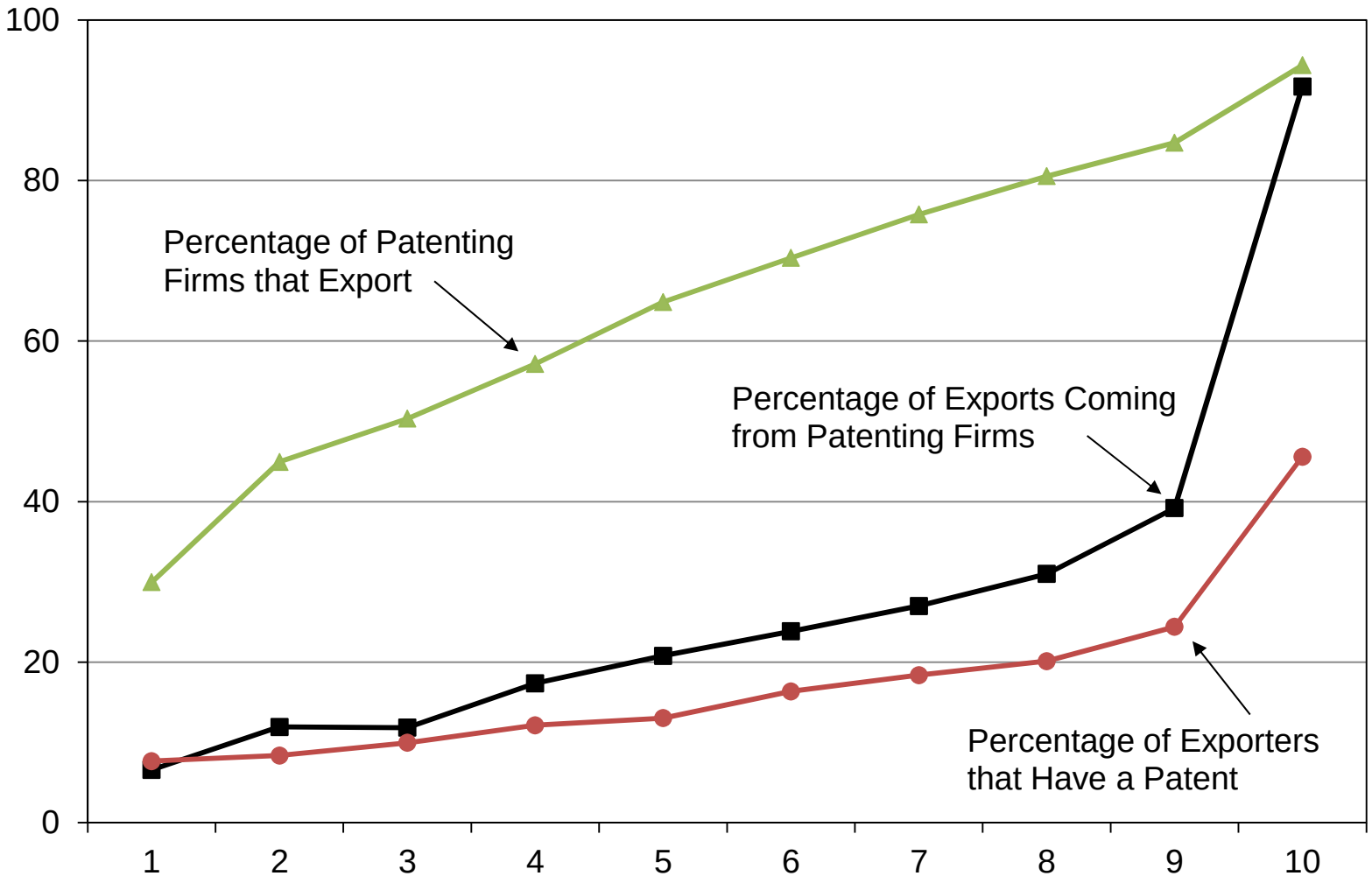


Figure 2: Exporting and Patenting Across Industries

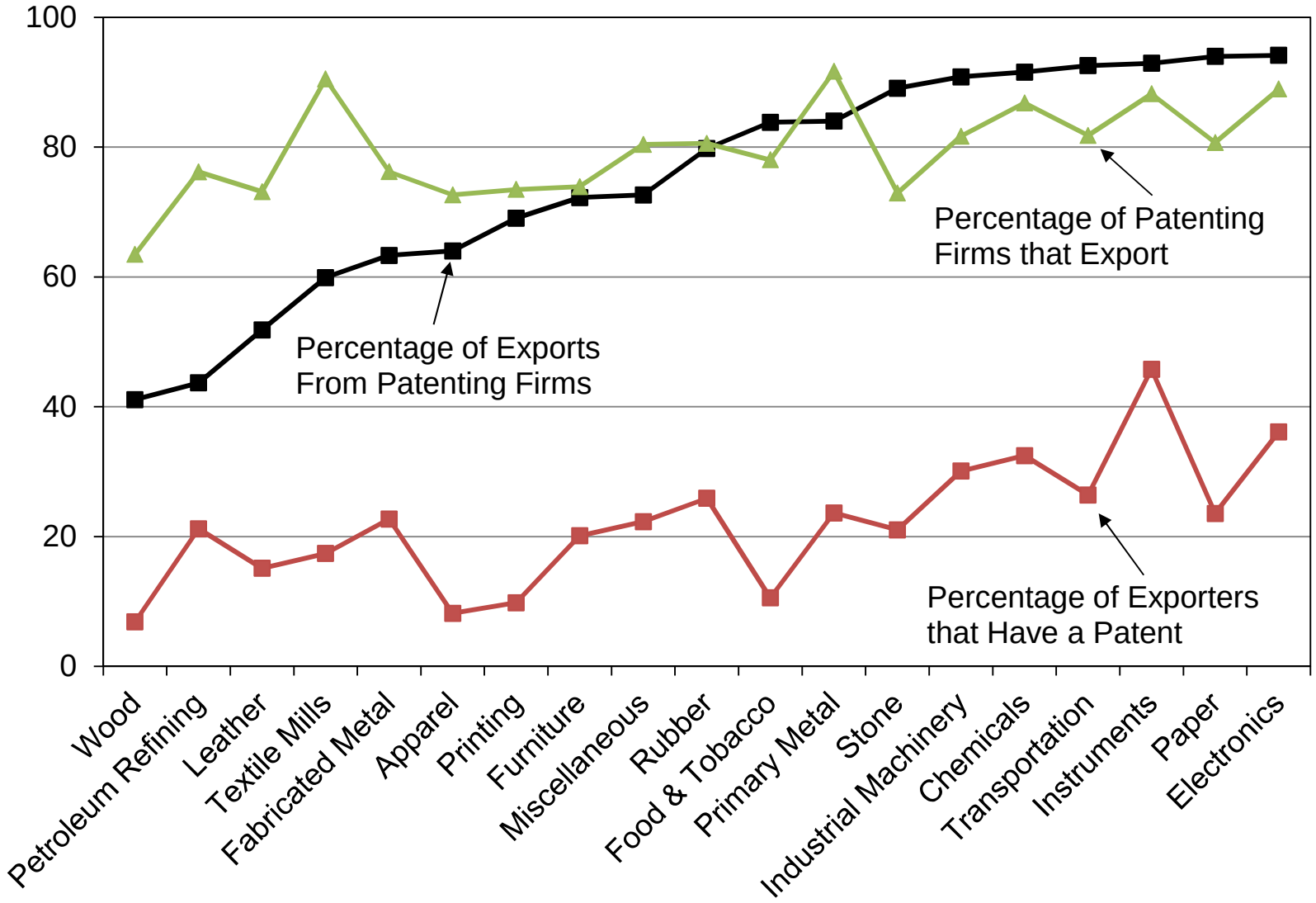


Figure 3: Exports From Patenting Firms

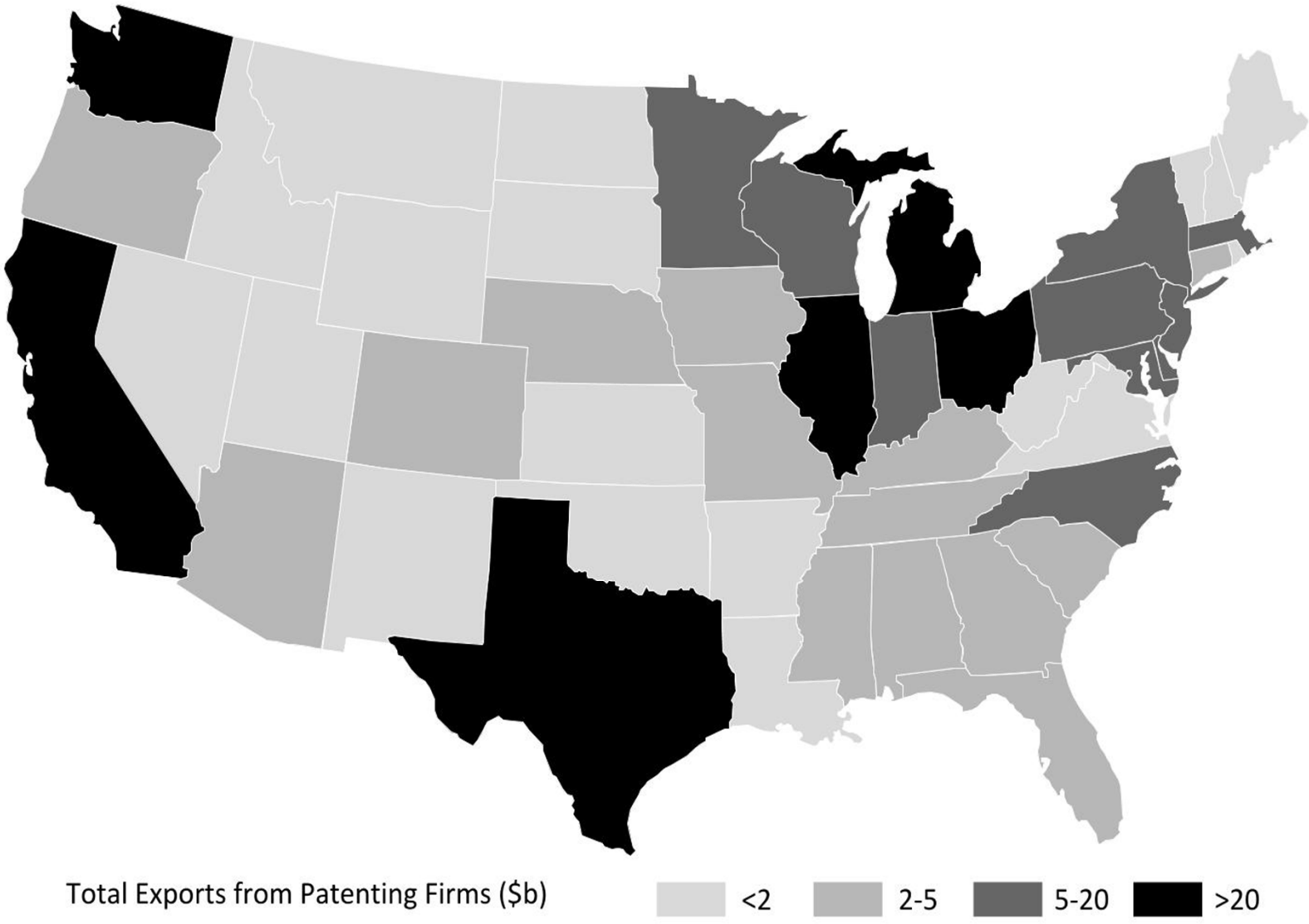


Figure 4: Percentage of Exports From Patenting Firms

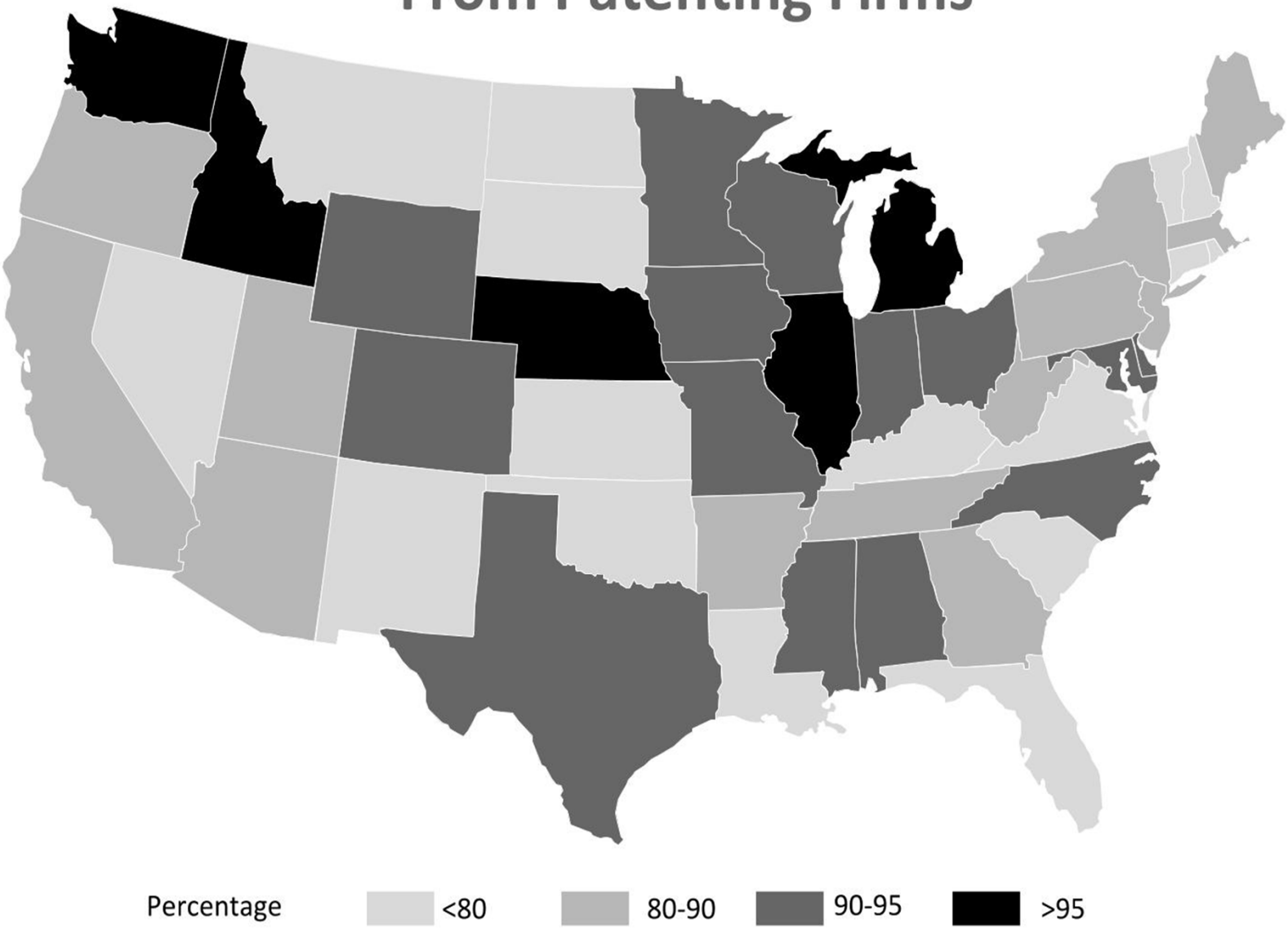


Figure 5: Exporting Around The Date Of A Firm's First Patent

